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Sample Question Set - Unit 6

Course Code: CSE276
Course Title: ARTIFICIAL INTELLIGENCE FOUNDATIONS
Course Outcome: CO6
Paper: Sample Multiple Choice Questions
Maximum Marks: 40 x 1 = 40

Unit: 6
Unit Title: AI Development Ecosystem and Future Trends
Unit Mapped RBT: L3 (Apply)
Total Questions: 40
Suggested Duration: 60 Minutes

Answer Grid (to be filled by the student)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1		Q11		Q21		Q31	
Q2		Q12		Q22		Q32	
Q3		Q13		Q23		Q33	
Q4		Q14		Q24		Q34	
Q5		Q15		Q25		Q35	
Q6		Q16		Q26		Q36	
Q7		Q17		Q27		Q37	
Q8		Q18		Q28		Q38	
Q9		Q19		Q29		Q39	
Q10		Q20		Q30		Q40	

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Name the Python library used for numerical arrays and vectorised arithmetic.

A) NumPy
B) Matplotlib
C) LangChain
D) Ollama

Q2. Identify the deep learning framework originally developed at Google.

A) PyTorch
B) TensorFlow
C) scikit-learn
D) Hugging Face

Q3. State what Ollama is used for.

A) Publishing datasets for other people to use
B) Storing embeddings for similarity search
C) Running open models locally on your own machine
D) Drawing charts from a table of results

Q4. Name the tool used to connect a language model together with retrieval and other components into one pipeline.

A) NumPy
B) TensorFlow
C) Matplotlib
D) LangChain

Q5. Identify the similarity measure most commonly used inside a vector database.

A) Manhattan distance
B) Cosine similarity
C) Hamming distance
D) Edit distance

Q6. State the Python library that provides the classical machine learning algorithms.

A) TensorFlow
B) Ollama
C) LangChain
D) scikit-learn

Q7. Explain what an approximate nearest neighbour index gives up in exchange for its speed.

A) It may occasionally miss the very closest match
B) It may return results in the wrong language
C) It may store the vectors in the wrong order
D) It may require the model to be retrained

Q8. Distinguish TensorFlow and PyTorch from scikit-learn by what each is used for.

A) The first two build classical models, the third networks
B) All three of them build only neural networks
C) The first two build neural networks, the third classical models
D) All three of them build only classical models

Q9. Describe what a virtual environment contributes to a Python project.

A) It speeds up the training of the model
B) It encrypts the data the project uses
C) It isolates the project's library versions from others
D) It converts the code into another language

Q10. Interpret the following requirement and name the component it calls for: an assistant must run with no data leaving the organisation's premises.

A) An open model served locally
B) A hosted commercial model over an API
C) A public dataset from a model hub
D) A shared vector store in the cloud

Q11. Predict what happens when the passages used for retrieval are made far too large.

A) The vector store refuses to accept them
B) The embedding model produces shorter vectors
C) The retrieval becomes exact rather than approximate
D) Each retrieved passage carries much irrelevant text

Q12. Recognize the stage described: a held-out set is consulted once in order to estimate how the system will behave in service.

A) Data collection
B) Evaluation before release
C) Problem definition
D) Monitoring after release

Q13. Explain why the embedding dimension must match between the stored vectors and the query.

A) The database will silently pad the shorter vector
B) The model will retrain itself to fit the store
C) The dimension has no bearing on the search
D) Similarity cannot be computed between vectors of different lengths

Q14. Summarize what sustainable Artificial Intelligence is concerned with.

A) The length of time a model remains accurate
B) The number of users a system can support
C) The energy and hardware that training and serving consume
D) The licence under which a model is released

Q15. Distinguish a model card from the model weights themselves.

A) The weights document the model, the card is the model
B) The card documents the model, the weights are the model
C) Both of the two are documentation of the model
D) Both of the two are numeric parameter files

Q16. Explain why the Artificial Intelligence development workflow is drawn as a loop rather than a straight line.

A) Monitoring sends the project back to earlier stages
B) Every stage has to be repeated a fixed number of times
C) The stages can be carried out in any order at all
D) The workflow has no defined end point of any kind

Q17. Describe what LangChain adds to a plain call to a language model.

A) A faster implementation of the model itself
B) Orchestration of prompts, retrieval and tool calls
C) A larger context window for the model
D) A guarantee that answers are factually correct

Q18. Predict the difficulty of reproducing a result when the versions of the libraries used were never recorded.

A) A later version may change the behaviour silently
B) The code will refuse to run at all
C) The data will be deleted automatically
D) The model weights will change by themselves

Q19. Explain why reusing a pre-trained model is more sustainable than training an equivalent one from the beginning.

A) Pre-trained models are stored in fewer bytes
B) Pre-trained models need no hardware to run
C) Pre-trained models require no evaluation
D) The largest share of the energy has already been spent once

Q20. Interpret the claim that a system has reached Artificial General Intelligence because it writes fluent essays.

A) The claim is correct, since writing requires reasoning
B) The claim is correct, since essays cover many subjects
C) Fluency in one task is not breadth across all tasks

D) The claim cannot be judged without the source code

Q21. Solve the following: 20,000 passages are embedded as vectors of 384 numbers, each stored in 4 bytes. Determine the memory the vectors occupy.

A) About 30.7 MB
B) About 7.7 MB
C) About 76.8 MB
D) About 3.1 MB

Q22. Compute the monthly cost of 15,000 queries charged at 0.004 dollars each.

A) 6 dollars
B) 600 dollars
C) 60 dollars
D) 150 dollars

Q23. Apply the development workflow and determine the stage at which the requirement and the success criteria are agreed.

A) Data preparation
B) Problem definition
C) Evaluation
D) Monitoring

Q24. Use the idea of a vector database to determine what is stored when 800 documents are added to a retrieval assistant.

A) One embedding for each whole document only
B) The complete text of every document alone
C) The parameters of the language model itself
D) One embedding per passage, with its source recorded

Q25. Examine a retrieval assistant that returns passages on entirely the wrong subject, and determine what to check first.

A) The temperature setting of the language model
B) The number of parameters the model holds
C) The passages, the embedding model and the index
D) The programming language of the application

Q26. Compare running a seven-billion-parameter model on a laptop with calling a hosted service, on latency and on cost.

A) Local costs per query and is usually faster
B) The two are identical on both latency and cost
C) Hosted costs nothing per query and is slower
D) Local costs nothing per query but is usually slower

Q27. Solve the following: a model requires 14 GB of memory and the available accelerator has 8 GB. Determine the appropriate action.

A) Quantise the model or choose a smaller one
B) Increase the temperature setting
C) Reduce the size of the context window
D) Add more documents to the vector store

Q28. Apply Hugging Face to a requirement for a pre-trained model and determine what it supplies.

A) The hardware on which the model will run
B) The weights, the model card and the code to load them
C) A guarantee that the model is free of bias
D) The private data of the organisation using it

Q29. Use the difference between exact and approximate search to determine which suits a store of ten million vectors.

A) Exact search, since it is always faster
B) Neither, since ten million vectors cannot be stored
C) Either, since the two cost the same at this size
D) Approximate search, since exact search is too slow

Q30. Compute the number of vectors held when 500 documents are each split into 20 passages.

A) 500
B) 10,000
C) 520
D) 20

Q31. Examine a deployment whose answers grow slower as the document store expands, and determine the remedy.

A) Reduce the number of documents in the store
B) Raise the temperature of the language model
C) Build a vector index in place of the linear scan
D) Retrain the language model on the documents

Q32. Apply version pinning to a project and determine what it prevents.

A) A silent change in behaviour after a library update
B) A rise in the cost of each hosted query
C) A leak of personal data from the vector store
D) A fall in the accuracy of the embedding model

Q33. Use the development workflow to determine what should be done immediately before a system is released.

A) Evaluate on the held-out set and record the result
B) Collect a further batch of training data
C) Increase the number of model parameters

D) Delete the documentation of the project

Q34. Compare a model card with an evaluation report on what each tells a prospective user.

A) The report states intended use, the card the performance

B) Both of the two state only the measured performance

C) Both of the two state only the intended use

D) The card states intended use and limits, the report the measured performance

Q35. Solve the following: an embedding model produces 768 numbers for each passage, stored at 4 bytes each. Determine the memory for 2,000 passages.

A) About 1.5 MB

B) About 6.1 MB

C) About 24.6 MB

D) About 0.8 MB

Q36. Apply the idea of sustainable Artificial Intelligence and determine which change most reduces the energy used in serving a model.

A) Adding more documents to the store

B) Raising the temperature of the model

C) Serving a smaller or quantised model

D) Writing a longer system prompt

Q37. Examine a project that can no longer reproduce a result obtained last month, and determine the most likely cause.

A) The evaluation set was too large to store

B) The vector store used cosine similarity

C) The library versions or the random seed were not recorded

D) The model card was written after release

Q38. Use the definition of Artificial General Intelligence to judge a system that translates fluently between sixty languages.

A) It qualifies, since sixty languages is very broad

B) It remains narrow, however impressive its range

C) It qualifies, since translation requires understanding

D) It cannot be judged without seeing its parameters

Q39. Demonstrate how an assistant grounded in internal documents is assembled from the tools of the ecosystem.

A) A single language model fine-tuned on every document

B) A relational database queried with exact keywords

C) A rule-based expert system with a knowledge base

D) An embedding model and vector store for retrieval, a served model to answer, and an orchestrator between them

Q40. Relate the number of stored passages to the need for an index rather than a straightforward comparison against every vector.

A) The larger the store, the greater the need for an index

B) The larger the store, the less the need for an index

C) The size of the store has no bearing on the need

D) An index is required even for a single passage

----- End of Sample Question Set -----